

# Introduction to Computing (CS 101)

## Review on Conditional Statements

**T. Venkatesh & John Jose**



**Department of Computer Science & Engineering**  
**Indian Institute of Technology Guwahati**

# Selection: the if-else statement

```
□ if ( condition )
```

```
□ {
```

```
□ statement(s)
```

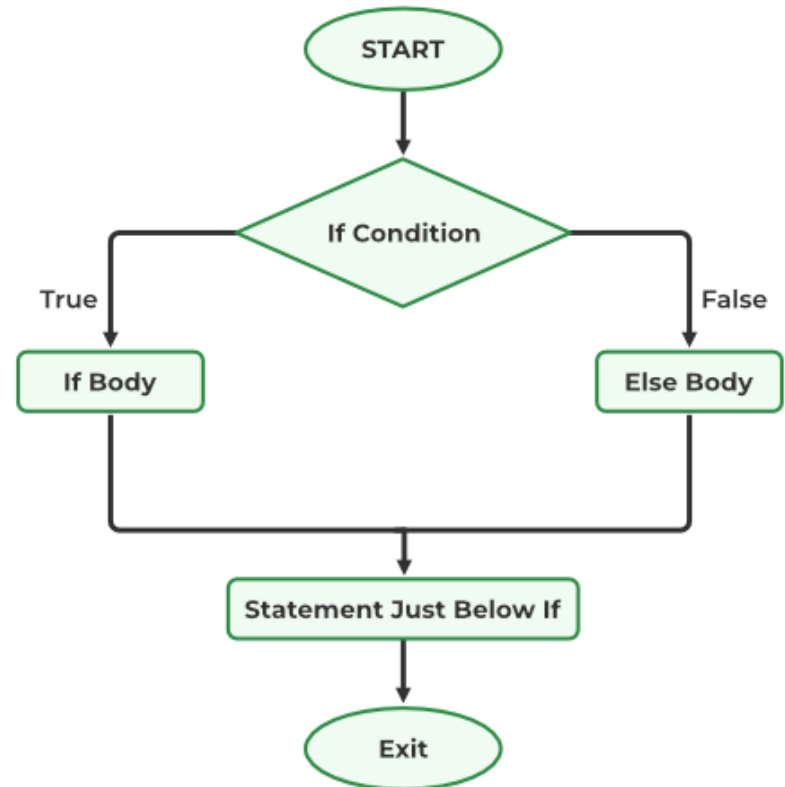
```
□ }
```

```
□ else
```

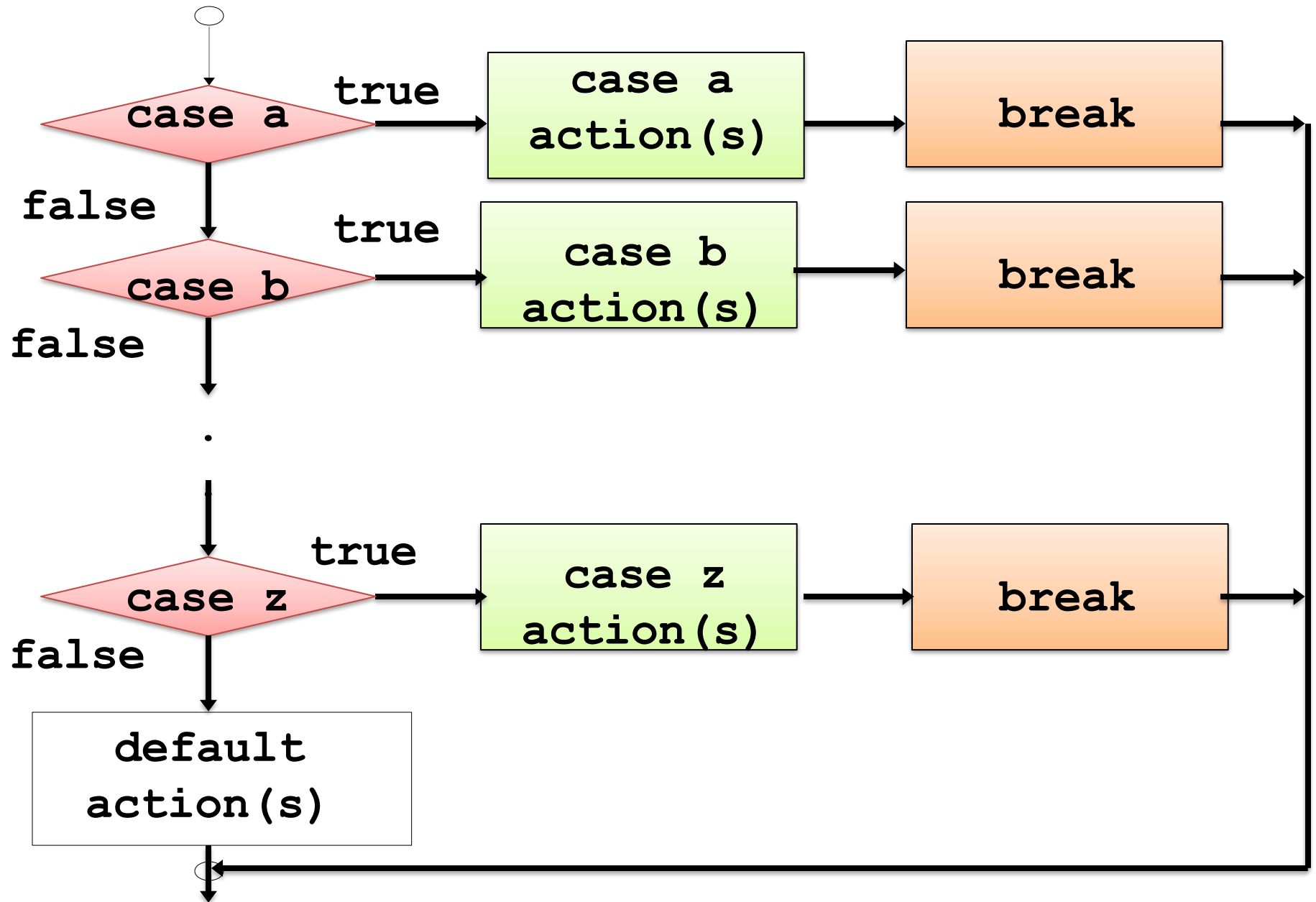
```
□ {
```

```
□ statement(s)
```

```
□ }
```

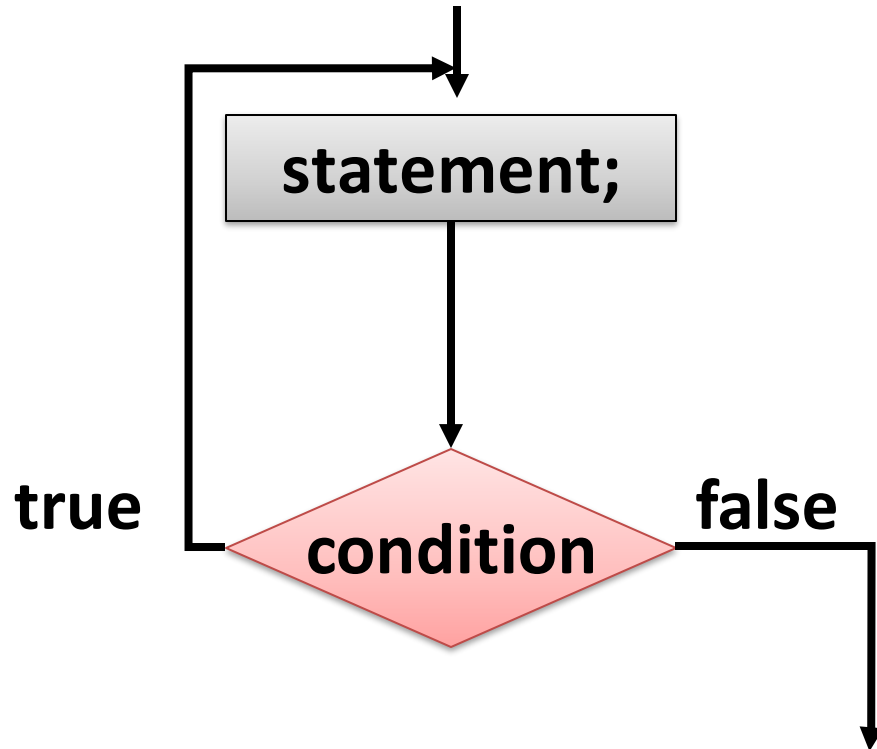


# Flowchart of Switch Statement

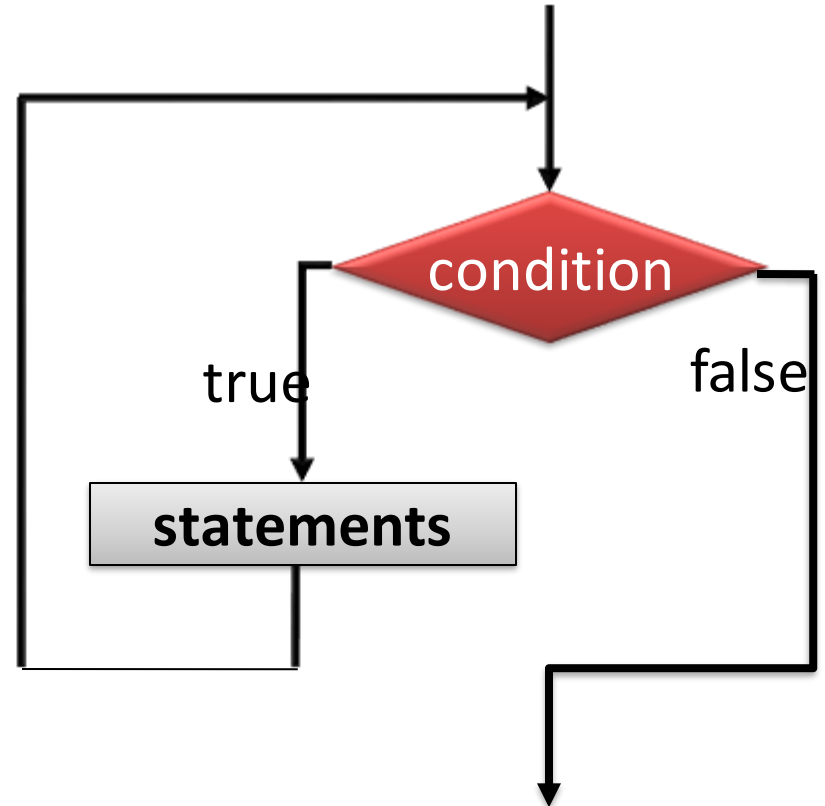


# Do while Vs While statements

do while loop



while loop



# Differences between do-while and while loops

| while                                                              | do-while                                                            |
|--------------------------------------------------------------------|---------------------------------------------------------------------|
| Condition is checked first before executing statements             | Statements are executed at least once before condition is checked   |
| Program uses<br><code>while(condition) { statement(s); }</code>    | Program uses<br><code>do { statement(s); } while(condition);</code> |
| while loop is entry controlled                                     | do-while loop is exit controlled                                    |
| If there is a single statement, brackets are not required.         | Brackets are always required.                                       |
| Variable in condition is initialized before the execution of loop. | variable may be initialized before or within the loop.              |

# Practice Program-1

Write a Program to check a given year is leap year or not.

```
int main() {  
    int year;  
    printf("Enter a year: "); // Input year  
    scanf("%d", &year);  
    // Check if the year is a leap year  
    if (year % 4 == 0) {  
        printf("%d is a leap year.\n", year);  
    } else {  
        printf("%d is not a leap year.\n", year);  
    } // end else  
    return 0;  
}
```

**Input:**

Enter a Year: 2000

**Output:**

2000 is a leap year.

# Practice Program-2

Write a program that takes a number as input from the user and performs one of the following operations based on the number entered:

If the number is 1, print "You entered 1".

If the number is 2, print "You entered 2".

If the number is 3, print "You entered 3".

If the number is anything else, print "Invalid input".

# Practice Program-2

```
int main() {  
    int choice;  
    printf("Enter a number (1, 2, or 3): ");  
    scanf("%d", &choice);  
    switch(choice) {  
        case 1:  
            printf("You entered 1.\n");  
            break;  
        case 2:  
            printf("You entered 2.\n");  
            break;  
        case 3:  
            printf("You entered 3.\n");  
            break;  
        default:  
            printf("Invalid input.\n");  
    }  
    return 0;  
}
```

Enter a Number (1, 2, or 3): 9

Invalid input.



# Practice Program-3

Write a Program using “Nested If-Else” statement which will do the following tasks.

1. It'll take as input of the name (char) & marks of 3 persons.
2. Then it'll calculate who is getting the highest marks.
3. Finally it'll print the name & marks of the 3 persons in decreasing order also by mentioning their respective ranks.

## Expected Output

Enter name and marks of person 1: R 90

Enter name and marks of person 2: A 88

Enter name and marks of person 3: D 95

| Rank | Name | Marks |
|------|------|-------|
|------|------|-------|

|   |   |    |
|---|---|----|
| 1 | D | 95 |
|---|---|----|

|   |   |    |
|---|---|----|
| 2 | R | 90 |
|---|---|----|

|   |   |    |
|---|---|----|
| 3 | A | 88 |
|---|---|----|

# Practice Program-3

```
int main() {
    char name1, name2, name3; // use single letter name
    int marks1, marks2, marks3;
    printf("Enter name and marks of person 1: ");
    scanf("%c %d", name1, &marks1);
    printf("Enter name and marks of person 2: ");
    scanf("%c %d", name2, &marks2);
    printf("Enter name and marks of person 3: ");
    scanf("%c %d", name3, &marks3);
    printf("Rank\tName\tMarks\n");
    if (marks1 >= marks2 && marks1 >= marks3) {
        printf("1\t%c\t%d\n", name1, marks1);
        if (marks2 >= marks3) {
            printf("2\t%c\t%d\n", name2, marks2);
            printf("3\t%c\t%d\n", name3, marks3);
        } else {
            printf("2\t%c\t%d\n", name3, marks3);
            printf("3\t%c\t%d\n", name2, marks2);
        }
    }
    // end if
}
```

```
else if (marks2 >= marks1 && marks2 >= marks3) {
    printf ("1\t%c\t%d\n", name2, marks2);
    if (marks1 >= marks3) {
        printf("2\t%c\t%d\n", name1, marks1);
        printf("3\t%c\t%d\n", name3, marks3);
    } else {
        printf("2\t%c\t%d\n", name3, marks3);
        printf("3\t%c\t%d\n", name1, marks1);
    }
} // end else if
else {
    printf ("1\t%c\t%d\n", name3, marks3);
    if (marks1 >= marks2) {
        printf("2\t%c\t%d\n", name1, marks1);
        printf("3\t%c\t%d\n", name2, marks2);
    } else {
        printf("2\t%c\t%d\n", name2, marks2);
        printf("3\t%c\t%d\n", name1, marks1);
    }
} // end else
return 0;
} // end main
```

# Practice Program-3

## Input & Output:

Enter name and marks of person 1: R 95

Enter name and marks of person 2: A 90

Enter name and marks of person 3: D 88

| Rank | Name | Marks |
|------|------|-------|
|------|------|-------|

|   |   |    |
|---|---|----|
| 1 | R | 95 |
|---|---|----|

|   |   |    |
|---|---|----|
| 2 | A | 90 |
|---|---|----|

|   |   |    |
|---|---|----|
| 3 | D | 88 |
|---|---|----|

Enter name and marks of person 1: R 90

Enter name and marks of person 2: A 88

Enter name and marks of person 3: D 95

| Rank | Name | Marks |
|------|------|-------|
|------|------|-------|

|   |   |    |
|---|---|----|
| 1 | D | 95 |
|---|---|----|

|   |   |    |
|---|---|----|
| 2 | R | 90 |
|---|---|----|

|   |   |    |
|---|---|----|
| 3 | A | 88 |
|---|---|----|

# Practice Program-4

Write a Program to calculate the sum of numbers from “m” to “n” by using “While” & “Do-While” Loop where “m” & “n” are two user defined integer input numbers.

# Practice Program-4

```
int main() {
    int m, n, p = 0, sum = 0;
    // Input values for m and n
    printf("Enter the value of m: ");
    scanf("%d", &m);
    p = m;
    printf("Enter the value of n: ");
    scanf("%d", &n);
    // Calculate the sum of numbers from m to n
    while (m <= n) { // will process only if m<=n
        sum += m;
        m++;
    }
    // Print the sum
    printf("The sum of numbers from %d to %d is: %d\n",
        p, n, sum);

    return 0;
}
```

**Input:** Enter the value of m: 2  
Enter the value of n: 10

**Output:**  
The sum of numbers from 2 to 10 is: 54

# Practice Program-5

Write a Program to print a Multiplication Table by using “while” and “do while” loop

```
int main() {  
    int num, i = 1;  
    // Input the number for the multiplication table  
    printf("Enter the number : ");  
    scanf("%d", &num);  
  
    // Print the multiplication table using a while loop  
    printf("Multiplication table of %d:\n", num);  
    while (i <= 10) {  
        printf("%d * %d = %d\n", num, i, num * i);  
        i++;  
    }  
    return 0;  
}
```

# Practice Program-5

## Multiplication Table by using “while” loop

```
int main() {
    int num, i = 1;
    // Input the number for the multiplication table
    printf("Enter the number : ");
    scanf("%d", &num);

    // Print the mult. table using a while loop
    printf("Multiplication table of %d:\n", num);
    while (i <= 10) {
        printf("%d * %d = %d\n", num, i, num * i);
        i++;
    }
    return 0;
}
```



# Practice Program-5

## Multiplication Table by using “do- while” loop

```
int main() {  
    int num, i = 1;  
    // Input the number for the multiplication table  
    printf("Enter the number : ");  
    scanf("%d", &num);  
  
    // Print the mult. table using a do-while loop  
    printf("Multiplication table of %d:\n", num);  
    do {  
        printf("%d * %d = %d\n", num, i, num * i);  
        i++;  
    } while (i <= 10);  
    return 0;  
}
```

# Practice Program-5

## Input:

Enter the number : 5

## Output:

Multiplication table of 5:

$$5 * 1 = 5$$

$$5 * 2 = 10$$

$$5 * 3 = 15$$

$$5 * 4 = 20$$

$$5 * 5 = 25$$

$$5 * 6 = 30$$

$$5 * 7 = 35$$

$$5 * 8 = 40$$

$$5 * 9 = 45$$

$$5 * 10 = 50$$

# Practice Program-6

Write a Program using “do-while” and “while” loops for printing a pattern mentioned below.

```
*  
* *  
* * *  
* * * *  
* * * * *
```

The number of rows will be defined by user input.

# Practice Program-6

## Pattern Printing using “do- while” loop

```
int main() {  
    int rows, i = 0, j=0;  
    // Input the number of rows from the user  
    printf("Enter the number of rows: ");  
    scanf("%d", &rows);  
    // Loop to print the star pattern  
    do {  
        j = 0;  
        do {  
            printf("* ");  
            j++;  
        } while (j <= i);  
        printf("\n");  
        i++;  
    } while (i < rows);  
    return 0;  
}
```

```
*  
* *  
* * *  
* * * *  
* * * * *
```

# Practice Program-6

## Pattern Printing using “while” loop

```
int main() {  
    int rows, i = 0, j=0;  
    // Input the number of rows from the user  
    printf("Enter the number of rows: ");  
    scanf("%d", &rows);  
    // Loop to print the star pattern  
    while (i < rows) {  
        j = 0;  
        while (j <= i) {  
            printf("* ");  
            j++;  
        }  
        printf("\n");  
        i++;  
    }  
    return 0;  
}
```

```
*  
* *  
* * *  
* * * *  
* * * * *
```

# Practice Program-7

Write a Program to print a given character is either vowel or consonant by using “while” Loop and “switch case” conditions. The program has to terminate if the user Input is “x”.

# Practice Program-7

```
int main() {  
    char ch;  
    while (1) {  
        // Input character from the user  
        printf("Enter a character (x to exit): ");  
        scanf(" %c", &ch);  
  
        // Check if the input character is 'x' to exit  
        the program  
        if (ch == 'x') {  
            printf("Exiting the program...\n");  
            break;  
        }  
    }  
}
```

# Practice Program-7

```
// Switch-case statement to determine whether
the character is a vowel or consonant
switch(ch) {
    case 'a':{printf("%c is a vowel.\n", ch); break;}
    case 'e':{printf("%c is a vowel.\n", ch); break;}
    case 'i':{printf("%c is a vowel.\n", ch); break;}
    case 'o':{printf("%c is a vowel.\n", ch); break;}
    case 'u':{printf("%c is a vowel.\n", ch); break;}
    case 'A':{printf("%c is a vowel.\n", ch); break;}
    case 'E':{printf("%c is a vowel.\n", ch); break;}
    case 'I':{printf("%c is a vowel.\n", ch); break;}
    case 'O':{printf("%c is a vowel.\n", ch); break;}
    case 'U':{printf("%c is a vowel.\n", ch); break;}
    default: {printf("%c is a consonant.\n", ch);}
        } // end switch
} // end while

return 0;
} // end main
```



# Practice Program-7

## Input & Output:

Enter a character (x to exit): p

p is a consonant.

Enter a character (x to exit): a

a is a vowel.

Enter a character (x to exit): R

R is a consonant.

Enter a character (x to exit): H

H is a consonant.

Enter a character (x to exit): y

y is a consonant.

Enter a character (x to exit): O

O is a vowel.

Enter a character (x to exit): u

u is a vowel.

Enter a character (x to exit): x

Exiting the program...

**Thanks**